



**Distributed Operating Systems and Algorithms | 6029 (20) | 001
(16970)
2026 Fall**

Course Information

Modality: In Person

Meeting Day(s): M W F

Meeting Time(s): 10:10AM

Meeting Location: WEST

Instructor Information

- **Name:** Nitin
Instructor Title: Assoc Professor - Educator (F2)
Instructor Qualifications: B.E.(Comp.Sc.Engg.) [Hons.],
M.E.(Soft.Engg.), Ph.D.(Comp.Sc.Engg.), Postdoc.(Info.Syst.),
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Instructor Contact Information

Office Location: 812C

Office Hours: MWF: 12:15-1:15

Office Hours Location: James A. Rhodes Hall

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Teaching Assistants and/or Peer Educators

Name: Nitin Nitin

Email: NITINFU@UCMAIL.UC.EDU

Course Description

Fundamental concepts of contemporary distributed operating systems. Topics include Fundamental Concepts Distributed Processes Distributed Resources Distributed Algorithms Privacy and Security Embedded Systems IoT Blockchain and Cloud Computing. - Prerequisite Definition: To take this course you must: Be enrolled in the following Program 20MAS. Be enrolled in the following Plan CSE-PHD.

Course Overview

Goals:

The goal of this course is to enable students to comprehend the function, design, and integration of contemporary distributed operating systems, real-time operating systems, and embedded systems, evaluate different operating system algorithms, and apply the knowledge of multi-process programming to efficiently utilize services provided by an operating system.

Course Learning Objectives:

Students will:

1. Understand the principles, architectures, algorithms, and programming models used in distributed systems. {1, 3}
2. Understand how independent network and machine failure conspire to make reliable distributed systems difficult to achieve. {1, 2, 6}
3. Articulate and understand the trade-offs between different well-known data consistency models. {1, 4, 6, 7}

4. Be able to design a distributed system for a specified level of fault tolerance and understand how to reason about the execution of a distributed system. {1, 2}
5. Understand the basic concepts of how to validate a running distributed system and apply cutting edge tools to investigate the behavior of systems under failure. {1, 3, 4, 6}
6. Understand practical design considerations for some modern distributed systems frameworks, and their limitations and recognize cases when distribution is not necessary. {1, 2, 3}
7. Understand the state-of-the-art distributed systems, such as Google File System and understand the design and implement sample distributed systems. {1, 2, 3, 4, 5}
8. Understand real-time operating system (RTOS) and the types of RTOS and to learn various approaches to real-time scheduling algorithms. {1, 2}
9. Understanding of Access matrices, hardening and cryptographic techniques. {1, 2, 6}
10. Understand the concept of embedded system, microcontroller, different components of microcontroller and their interactions with IoT devices and Blockchains including Cloud Computing. {1, 2, 4, 7}

ABET outcomes EAC 2021-2022:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Course Format

The following contents will be covered:

- Fundamental Concepts
 - § Transparency,
 - § Service, and
 - § Coordination
- Distributed Processes
 - § Synchronization,
 - § Communication, and
 - § Scheduling
- Distributed Resources
 - § Files and
 - § Memory
- Distributed Algorithms
 - § Time and Global States in a Distributed System,
 - § Distributed Synchronization and Distributed election,
 - § Distributed Agreement Problem,
 - § Replicated Data Management, and
 - § Distributed Rollback and Recovery
- Privacy and Security

§ Distributed Access Matrix, and

§ Kerberos V & SHA 256/512

- Embedded Systems

§ RMS-EDF,

§ Fault Tolerant RTOS, and

§ Load Balancing

- IoT, Blockchain and Cloud Computing

§ Case Study on SCM by IBM

Course Learning Outcomes

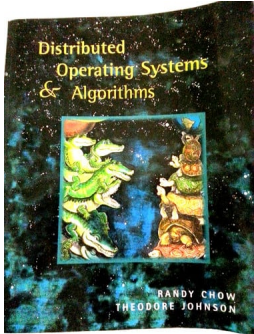
Illustrate the principles architectures and programming models used in distributed systems. Recognize how independent network and machine failure conspire to make reliable distributed systems difficult to achieve. Debate and understand the trade-offs between different well-known data consistency models. Design a distributed system for a specific level of fault tolerance and understand how to reason about the execution of a distributed system. Characterize the basic concepts of how to validate a running distributed system and apply cutting edge tools to investigate the behavior of systems under failure. Assess practical design considerations for some modern distributed systems frameworks and their limitations and recognize cases when distribution is not necessary. Research state-of-the-art distribution systems such as Google File System and understand the design and implement sample distributed systems. Evaluate real-time operating systems (RTOS) and the types of RTOS and learn various approaches to real-time scheduling algorithms.

Course Related Costs

Students must purchase a required text or reading material

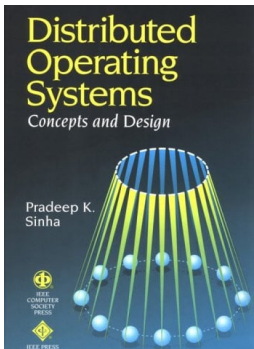
Required Course Materials

[University of Cincinnati Bookstore](#)

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Title: Distributed Operating Systems & Algorithms
ISBN: 9780201498387
Authors: Chow, Randy, Johnson, Theodore
Publisher: Addison-Wesley
Publication Date: 1997-03-28

Recommended Course Materials

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Title: Distributed Operating Systems: Concepts and Design
ISBN: 9780780311190
Authors: Sinha, Pradeep K.
Publisher: Wiley-IEEE Press
Publication Date: 1996-12-15
Edition: 1
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Title: Distributed Systems: Principles and Paradigms
ISBN: 9780130888938
Authors: Tanenbaum, Andrew S., Van Steen, Maarten
Publisher: Prentice Hall
Publication Date: 2002-01-15

Artificial Intelligence (AI) Guidelines

Appropriate to use for limited purposes – In this course, students are encouraged to use Generative AI Tools like ChatGPT to support their work. However, not all use cases of Generative AI are appropriate in this course. Students may use these tools to [BearcatsGPT]. Other uses beyond this list will be considered

academic misconduct. Please do not hesitate to ask if you are unsure if your specific application is appropriate in this course. Students should exercise caution with the output of Generative AI tools – they often generate incorrect statements despite attempts to steer them toward correctness. Students are responsible for verifying the accuracy of the outputs it provides.

What coursework can be completed with the help of generative AI tools?

Selected Assignments - There will be certain assignments in this course where it is appropriate for students to use Generative AI Tools to complete their work; however, it is expected that other assignments will constitute the student's own work. It will be explicitly communicated by the instructor and included in the instructions when students are allowed to explicitly use Generative AI Tools to complete an assignment. Using Generative AI Tools to complete assignments beyond the established use cases that are not designated as appropriate for AI assistance constitutes academic dishonesty.

AI Attribution

Citation and chat history/prompts are required -

A student should include the following statement in assignments to credit the use of a Generative AI Tool: "The author(s) would like to acknowledge the use of [BearcatsGPT], developed by [University of Cincinnati], in the preparation of this assignment. The [BearcatsGPT] was used in the following way(s) in this assignment: [e.g., brainstorming, grammatical correction, citation, which portion of the assignment]."

The student must also submit the chat history or prompts, whichever is most appropriate for the generative AI tool, as additional evidence of how these resources were used to create the submitted work. Sharing chats can be done easily in ChatGPT by using the Share chat feature. Additional information is provided at the following URL: <https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq>. Otherwise, students must submit the prompts they used when completing the assignment in a PDF file.

Course Calendar

- Week/Module: Week 1/ Module 1
Topic: Fundamental Concepts
Material: Book Chapter
- Week/Module: Week 3/ Module 2
Topic: Distributed Processes
Material: Book Chapter
- Week/Module: Week 6/ Module 3
Topic: Distributed Resources
Material: Book Chapter
- Week/Module: Week 8/ Module 4
Topic: Distributed Algorithms
Material: Book Chapter
- Week/Module: Week 11/ Module 5
Topic: Privacy and Security
Material: Book Chapter
- Week/Module: Week 12/ Module 6
Topic: Embedded Systems
Material: Book Chapter
- Week/Module: Week 14/ Module 7
Topic: IoT, Blockchain and Cloud Computing
Material: Book Chapter

Assessment/Activities and Grading Policy

Final numerical grades for the course are determined based on the following weights for assignments.

Assignment Weighting	
Assignment	Weighting
Homework	60%
Quizzes	0%
Lab assignments	0%
Exams	20%
Projects	15%
Reports	5%
Pre-class materials	0%
Discussion	0%

Assignment	Weighting
Participation	0%
Other	0%
Total	100%

Final letter grades for the course will be determined using your final numerical grade and the following ranges. Please note that the instructor reserves the right to modify the range for each letter grade. However, the ranges posted here provide the minimum score you must achieve to guarantee the grade indicated.

Numeric Range	
Grade	Range
A	92-100
A-	89-92
B+	86-89
B	82-86
B-	79-82
C+	76-79
C	72-76
C-	69-72
D+	66-69
D	62-66
D-	59-62
F	Below 59

Pass/Fail, Audit, and Withdrawal

- For information on Adding, Dropping, and Withdrawing from classes, please visit: [University of Cincinnati Office of Registrar](#)
- For information on Grading Scales and Definitions, please visit: [University of Cincinnati Grading Scales and Definitions](#)

Technology use during/for class

For technology resources such as the DTS Knowledge Base (KB) tutorial articles and the Help Desk, visit the [DTS Help Page](#). Currently the student text is blank.

Attendance Policy

Students at the University of Cincinnati are expected to attend classes in order to meet the learning objectives for the course. Students are expected to follow the class attendance policy as outlined by the instructor. Students participating in official University sponsored activities, where the student is representing the University, will meet with instructor related to absences due to university-sponsored activities. When such absences occur, students need to provide documentation of the event(s), and work with the professor to develop a written plan for completion of missed assignments or other course requirements with as much advance notice as possible. Any absence due to official University business may still count against the total number of allowable absences.

Faculty Attendance

Course Delivery Mode:

- This course is a F2F Course.
- This course can be found on Canvas (accessible via [canopy.uc.edu](https://canvas.uc.edu)). Assignments, reading materials, videos, etc. will be posted on the course site in Canvas. In difficult weather situation Live online lectures will be conducted using [Zoom]. Invitations that link to scheduled live online lectures will be posted on the course site in Canvas.

Contact Information/Office Hours: You should not hesitate to contact your instructor or come to office during office hours (See canopy for official office hours) if you have any problems, concerns, questions, or issues regarding anything related to this course. The quickest and possibly the best way to get a hold of your instructor is via Email.

In general, most of us have an “open door” policy. If our door is open, stop on in and say “hello” or ask a question. If our door is closed, please come back during office hours or later in the day. Please note that the office hours may be in the form of on-line if the pandemic keeps severe to protect all of us.

If you need to meet at certain time, shoot us an Email. In many cases, the problem can be corrected via Email.

Note: Please include “EECE DOS” in the Subject of any Email message so we can pick it out of the crowd.

Class Cancellation Policy

In the rare case that a class must be cancelled, or that a faculty member is absent, faculty will post an announcement on Canvas or email information to students. Faculty will attempt to communicate class cancelations with as much advance notice as possible. Students should be sure that their LMS email is current and valid to ensure emails are received.

For information on class cancellation due to emergency procedures follow the link below:

[Emergency Closing Procedure.](#)

Notice of Non-Discrimination

- For the University of Cincinnati's Notice of Non-Discrimination, please visit: [UC Notice of Non-Discrimination](#)

Student Religious Accommodations

- For information on the University's Religious Accommodations Policy, please visit: [Student Religious Accommodations Policy](#)

Accessibility Policy

If you have a physical, psychological, or cognitive disability (learning, ADD/ADHD, psychological, visual, hearing, physical, cognitive, medical condition, etc.), UC's Accessibility Resources and your college's Access Coordinator are here to ensure you receive reasonable accommodations.

I encourage you to contact [Accessibility Resources](#) to schedule a confidential meeting to discuss services and accommodations, or to begin by registering for accommodations. UC uses an online system called [UC Accommodate](#), where you'll be able to register and manage your accommodations (use your UC username and password for access).

Below are a few helpful UC Accommodate resources:

[UC Accommodate Student Portal Overview Links to an external site.](#)

[How to Schedule an Appointment in UC Accommodate Links to an external site.](#)

[How to Submit a Semester Request in UC Accommodate](#)

Academic Integrity and Learning Commons

For information on the University's Academic Misconduct Policies.

[Academic Misconduct | University of Cincinnati](#)

Learning Commons

The Learning Commons provides centralized academic support for all University of Cincinnati students by offering tutoring, coaching, writing support, and many additional programs that you can access via registered courses, appointments, or just dropping into the office. Learning Commons professionals and Peer Educators are available to help create flexible academic success programming whether you need some assistance with brainstorming ideas for your assignment, improving a draft, or managing your time. Check out the Learning Commons website to learn about available support and how to schedule appointments. You can also call the Learning Commons at 513-556-3244 or stop by the desk on the ground floor of French Hall West or Langsam Library at the Academic Writing Center (AWC).

Ethics and integrity are core values that should guide our conduct and decisions as members of the UC community and as engineering and technology professionals. Course assignments and tests are designed to have educational value and prepare you to be a competent and ethical professional. Therefore, material presented by you to satisfy course requirements is expected to be result

of your own original scholarly efforts, unless collaboration is explicitly allowed on a particular assignment.

University Rules, including the [Student Code of Conduct](#) and other documented policies of the department, college, and university related to academic integrity, will be enforced. Any violation of these regulations, including acts of plagiarism or cheating, will be dealt with on an individual basis according to the severity of the misconduct and can result in severe consequences, including potential dismissal from the college after the second offense.

Title IX

For information on the University's Title IX Policy click the link:

[Title IX Policy - Ethics, Compliance & Community Impact | University of Cincinnati](#)

Help for Student Victims & Student Survivors

The [Help for Student Victims & Student Survivors website](#) provides resources for victims or survivors of sexual harassment, including sexual assault, dating or domestic violence, gender-based harassment, or stalking. Staff are available 24-hours a day for confidential advice and assistance.

Bearcats Pantry and Resource Center

[The Bearcats Pantry and Resource Center](#) is available to all UC students. From free food to social services support, they provide a range of programs, services, and supplies. Each student may visit once per week and receive up to 20lbs of goods per visit. You will need to make a [PantrySOFT account](#) and present your physical Bearcat Card for entry. They also encourage you to bring a reusable bag when you visit.

Mental Health Resources

Mental Health counseling and a variety of resources are available to UC students through [Counseling & Psychological Services \(CAPS\)](#). The Crisis, Assessment,

Referral, Evaluation Team (“CARE Team”) responds to reports about students whose behavior is raising concerns within the University community. More information about the CARE Team can be found in the [University's Mental Health Assessment Policy](#).

UC Public Safety

The Department of Public Safety at the University of Cincinnati is committed to providing a safe campus environment for students, faculty, staff and visitors. Visit the [UC Public Safety website](#) to learn more about public safety and emergency response resources and services.

Inclusivity

This class is committed to the fundamental principles of academic freedom and human dignity. Diversity in all forms is something we welcome, we foster, and we prize. We believe that honest attempts to understand the perspectives of others facilitates learning, and we will strive to achieve this goal at all times. We strongly disavow discrimination -- including harassment -- on the basis of race, national or ethnic origin, religion, sex or gender identity, disability, age, sexual orientation, or veteran status. We expect that each of us will hold one another accountable for maintaining these ideals.

All are welcome and considered a valuable addition to the university community. You should consider my classroom as an inclusive and safe space to express your ideas and viewpoints. No discrimination is accepted or tolerated in this course. It is the goal for you to be successful and to thrive to your highest potential. [UC Notice of Non-Discrimination](#)

Special Statement on COVID-19

[UC's COVID-19 Public Health Webpage](#) contains the most up-to-date policies and practices regarding COVID-19 precautions at UC. Remember, sick and/or quarantined students should not come to class. Instead, please contact the instructor to discuss appropriate online accommodations.

Right to Modify the Syllabus

I reserve the right to update this syllabus as class needs arise. Be assured that I will communicate to you any changes to our schedule, syllabus or policies quickly and efficiently through [\[nitinfu@ucmail.uc.edu\]](mailto:nitinfu@ucmail.uc.edu).

Course Detailed Description

Detailed Description

Fundamental Concepts (Transparency, Service, and Coordination)

Chapter 1: introduces a classification of centralized operating system, network operating system, distributed operating system, and cooperative autonomous systems, using the key characteristics of virtuality, interoperability, transparency, and automaticity, respectively, for each system. It illustrates the evolution that led to the development of modern distributed operating systems and explains the emerging need for distributed software and the importance of distributed coordination algorithms.

Chapter 2: begins the discussion of distributed operating systems. It presents the concepts of transparency and services. Distributed systems and their underlying communication architectures are introduced. The chapter concludes with a list of major system design issues that establishes an order for the presentation of the subsequent chapters.

Distributed Processes (Synchronization, Communication, and Scheduling)

Chapter 3: describes concurrent processes and programming. It defines processes and threads and shows how their interaction can be modeled by using some fundamental concepts such as a graph, a logical clock, and the client and server model. Both shared memory and message passing for synchronization

and communication are addressed. They are presented along with the development of concurrent language constructs. A taxonomy of these language mechanisms and their implementation is given. This chapter presents an integrated view of synchronization and communication.

Chapter 4: extends the discussion of process interaction from synchronization to communication and to distributed process coordination using message passing communication. Three communication models, message passing (socket), request/reply(RPC), and transaction communication, are presented. A special emphasis is placed on group communication and coordination. Two classical distributed coordination problems, mutual exclusion and leader election using message passing interprocess communication, are introduced. These problems are further studied in Chapters 10 and 11 in Part II of the textbook. The chapter also includes a presentation of name service, an essential facility for communication in distributed systems.

Chapter 5: turns to the third process management issue, that of process scheduling. The effect of communication on both static and dynamic process scheduling is emphasized. The chapter describes distributed computation through dynamic redistribution of processes by using remote execution and process migration techniques. It also addresses several unique issues in real-time scheduling.

Distributed Resources (Files and Memory)

Chapter 6: discusses the distributed implementation of file systems, the first of the two important distributed resources: files and memory. It demonstrates the use of the concept of transparency and service in the design of distributed file systems. Two major implementation issues, data caching and file replication, are discussed in this chapter. The chapter also covers distributed transactions as part of the file service. Since management of replicated data touches upon both data and communication, two central issues in distributed systems, it is further detailed in Chapter 12.

Chapter 7: covers distributed shared memory systems that simulate a logical shared memory on a physically distributed memory system. The issues studied are coherence and consistency of data due to memory sharing. The chapter describes implementation strategies for different memory consistency requirements. It also shows the significance of the object-based data sharing models.

Chapter 8: addresses unique security issues in network and distributed environments. These issues are divided into two areas: authorization and authentication. Authorization includes the study of distributed access and flow control models. Authentication covers cryptography and its applications for mutual authentication and key distribution protocols. Implementations of some security features in modern systems are illustrated.

Part II of the textbook discusses distributed algorithms. The discussion is pragmatic and is intended to give the reader a solid understanding of common problems and solution techniques. The topics are organized in five chapters.

Distributed Algorithms

Chapter 9: introduces the concepts of time and global states in a distributed system. The fundamental problem of distributed algorithms is a lack of a global clock and a global state. Recent research on vector time and distributed predicates has developed unified models for thinking about distributed time and the distributed state. This chapter presents the concepts of causality, vector timestamps, and global states. The algorithms for implementing these concepts are presented. The connections between the different models are explored. Finally, a model for proving the correctness of distributed algorithms is presented.

Chapter 10: covers distributed synchronization and distributed election. While the distributed synchronization algorithms are not considered pragmatic, they

illustrate important algorithm design techniques. For example, voting algorithms for replicated data management are foreshadowed in Maekawa's algorithm, and the Chang-Singhal-Liu algorithm illustrates the ideas behind distributed shared memory (and distributed object) algorithms. The chapter concludes with algorithms for electing a computation leader. Election is a critical component of many systems. The invitation algorithm is a prototype for handling failures in an asynchronous system and foreshadows the group view maintenance algorithms of Chapter 12.

Chapter 11: discusses the abstract distributed agreement problem. First, Byzantine agreement is discussed. Next, the Fischer-Lynch-Paterson (FLP) result that no algorithm solves distributed agreement problems in an asynchronous system is covered in detail. This is the appropriate point to introduce the FLP result because the next chapter covers replicated data management and must solve distributed agreement in asynchronous systems. The FLP result leaves open three ways to achieve distributed agreement in an asynchronous system: hope that it happens, use relative agreement, or use a randomized algorithm. The chapter discusses these implications of the FLP result and concludes with some randomized agreement protocols.

Chapter 12: covers replicated data management. Since providing replicated servers reduces to replicating the state of the servers, this section also discusses the problems and concepts of replication. We cover three main approaches: the transaction approach, the reliable multicast approach, and the log propagation approach. The transaction approach includes discussion of two-phase commit, three-phase commit, one-copy serializability, voting, and dynamic voting protocols. The reliable multicast approach includes discussion of virtual synchrony, algorithms for implementing reliable and causal multicast, algorithms for totally ordered multicast, and consistent multicast group maintenance algorithms. The log propagation approach covers naive log propagation, epidemics, and causal log propagation. This chapter is the culmination of Part II of the text and draws together the results presented in previous chapters.

Chapter 13: covers distributed rollback and recovery. These techniques are critical for implementing fault-tolerant systems and are complementary to the replicated data management techniques of the previous chapter. By using the

theory developed in the previous chapters (especially Chapter 9), different rollback and recovery algorithms are presented in a unified manner and are related to algorithms discussed previously.

Student Code of Conduct

Students are expected to work individually for their Assignments. They may discuss strategies and pseudo code with other individuals but may not share any code with other individuals. If you get code/solution from an outside source such as the text or a web site, include a reference to that in your comments. Failure to include a reference to any code/solution from the text or from a website will be considered the same as copying code/solution from another student. Copying code/solution on a homework assignment will result in a 0 points for that assignment for all parties involved. Repeated infractions will result in an F in the course and referral to the college for potential additional sanctions.

In other words,

Students are to follow the University's Student Code of Conduct:

1. Do not plagiarize.
2. Do not misrepresent work completed by others as your own.
3. First-time violators will receive a zero on the exam or assignment. Repeat violators will receive a semester grade of 'F' and are referred to Judicial Affairs for Academic Misconduct.

Regrade Requests:

A request for a re-grade of a graded item must take place within a week after you receive the graded item. No re-grade requests will be honored after that time.

Policies and Expectations

1. All Students will need a laptop to complete this course.

2. Check UC email and Canopy daily for announcements, helpful tips, and updates.
3. Attend every class, arrive on time, arrive prepared (read the chapters and/or watch the assigned videos before class). Class attendance is expected and it is in your best interest.
4. The textbook or canopy may not cover all topics and details introduced during the lecture. However, Students are still responsible for this information.
5. Your instructor is to treat you with respect, answer your questions, be prepared to lecture, and be available outside of class for office hours or via Email. Email responses will be sent during standard business hours.
6. Do not disrupt the class, other students in the class, or the instructor.
7. Participate, keep up, and act professionally.
8. Use of computers during class is for coursework only.
9. No extra credit work will be available.
10. Report any special needs requirements to the instructor as soon as possible.
11. Email your instructor immediately, if you need to hand in an assignment late, or miss an exam or quiz.